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Is the dynamics of polystyrene films consistent with their glass transition temperature?

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Abstract. Experiments in the past 1.5 decades have found that the glass transition temperature of polymer films can be noticeably different from the bulk when the film thickness is decreased below ∼100 nm. On the other hand, many dynamic measurements have found results inconsistent with the observed change in the glass transition temperature. One frequently cited reason is that the dynamic properties being probed may not be directly related to the glass transition. Viscosity is a property traditionally used to characterize the dynamic slowing down occurring to a material at the glass transition. In this paper, we report experimental result showing that the viscosity of polystyrene films supported by oxide-coated silicon decreases with decreasing film thickness, consistent with the observed glass transition temperature of the films.

1 Introduction

Glass transition temperature, T_g, is an important engineering property of polymers because it is the temperature at which a material softens upon heating and loses its ability to hold its shape. Studies in the past 16 years have shown that the T_g of polymer films can be very different from the bulk. In the seminal work of Keddie et al. [1], the T_g of polystyrene (PS) films on silicon was first measured, and observed to decrease with decreasing film thickness when the film thickness was decreased below 100 nm. Since then, several other systems have been found to exhibit decreasing T_g with decreasing film thickness, including poly(methyl methacrylate) (PMMA) on gold [2], syndiotatic PMMA on silicon or aluminum [3], freely standing PS [4] and PMMA films [5], PS end-functionalized with fluoroalkylsilyl groups on silicon [6], as well as poly(α-methyl styrene) and polysulfone on silicon [7]. On the other hand, polymer films exhibiting increasing T_g with decreasing film thickness have also been reported. These include PMMA on silicon [2], isotatic PMMA on silicon or aluminum [3], PS on hydrogen-terminated silicon [8], and poly(2-vinylpyridine) on silicon [9]. Typically, the T_g of polymer films is determined by measuring the thermal expansion of the film during a temperature scan that covers the glass transition.

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More recently, new techniques are being introduced, which often also involve more stringent experimental conditions, such as ultra high vacuum and extensive sample pre-annealing. For reasons that are still under debate, many of these experiments did not find any shift in the T_g from the bulk value [10–13].

The most commonly used technique to measure the T_g of polymer films is ellipsometry [1]. In this technique, the ellipsometric angles, Δ and/or Ψ are monitored as a function of the sample temperature. Because Δ and Ψ are usually proportional to the polymer thickness, a plot of Δ or Ψ versus temperature is often taken to be a manifestation of the thermal expansion of the film. At the glass transition, the thermal expansion coefficient of the film changes discontinuously. This property has been exploited in the determination of the T_g of the films. Specifically, it is taken to be the temperature at which a plot of Δ or Ψ versus temperature exhibits a discontinuous change in the slope.

While the notion that the first derivative of a thermodynamic property, such as heat capacity and thermal expansion coefficient, changes discontinuously at the glass transition is commonplace, the glass transition is more appreciated - at least from practical point of view – as the process in which the viscosity of a material increases rapidly upon cooling past the T_g. The size of the increase is so large that the material changes from the liquid state to a solid-like glassy state within only a few °C of cooling past the T_g. The reason why the problem of thin film T_g is significant and has drawn widespread attention is its technological implication. With the results discussed above, if one makes a nanostructure out of a polymer, that structure may collapse even upon moderate heating from room temperature, which can substantially restrict how the polymer can be used in nanoscale applications.

Numerous experiments had been carried out to determine how the dynamics of polymer films may change as a result of the observed change in the T_g. Typically, these experiments measure the relaxation time, τ, as a function of the sample temperature, T. For bulk polymer, the τ-T as well as the viscosity-T dependence have been found to follow the Vogel-Fulcher-Tammann (VFT) relation [14,15]:

$$\tau = \tau_\infty \exp\left[\frac{A}{k_B(T - T_K)}\right], \tag{1}$$

where T_K is the Kauzmann temperature at which the VFT relation extrapolates to infinity, τ_∞ and A is, respectively, the relaxation time and activation energy of the system when $T \gg T_K$ and $T \gg A/k_B$. There is no well-defined relation between the T_g and the VFT relation. But since T_g is conventionally measured as the sample temperature is swept at one to several °C per minute (as for instance in differential scanning calorimetry and dilatometry) and the width of the glass transition is typically several °C, the T_g is commonly regarded to be the temperature at which the relaxation time is ∼1 to ∼10 s. In a similar spirit, Donth [16] suggested that T_g is the temperature at which the viscosity increases to an arbitrarily large value of $\sim 10^{12}$ Pa·s whereupon the material does not flow in practical times. Less straightforwardly, the T_g may sometimes be perceived as being related to the T_K. For polystyrene, the T_g has been found to be always about 50 °C above the T_K irrespective of the molecular weight of the polymer [17].

Based on the above discussions, two changes in the VFT relation and their combinations can be consistent with a reduction in the T_g. One is a reduction in the divergence of the relaxation time or viscosity with decreasing temperature (which may, for example, result from a reduction in the T_K as illustrated by the dashed line in Fig. 1.) The other is a reduction in the relaxation time or viscosity over the entire temperature range (which may result from a reduction in τ_∞ as illustrated by the dotted line in Fig. 1). In the actual experiments, however, results that are far more complicated and sometimes even contradictory have been found.

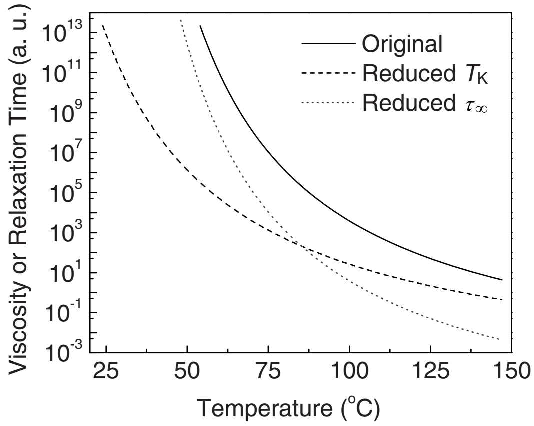

Fig. 1. An illustration of how a reduction in T_K or τ_∞ may affect the VFT relation. The original VFT curve is represented by the solid line. The curve after the T_K is reduced is represented by the dashed line; that after τ_∞ is reduced is represented by the dotted line. Note that °C rather than K is used here as the unit for temperature. This choice does not affect the plot qualitative as only the difference in temperature enters the equation for the VFT relation (Eq. (1)).

Organization of this paper is as follows. We first review several controversial results concerning the T_g and dynamics of polymer films, notably those determined by ac dielectric spectroscopy. Then we discuss the result we obtained by monitoring the dewetting of polystyrene films supported by oxide-coated silicon. We find that the viscosity of the films deduced from the dewetting experiment decreases monotonically with decreasing film thickness, consistent with the T_g measured by ellipsometry.

2 Methods

2.1 Sample preparation

The substrates used in this experiment are single crystal (001) silicon wafers covered by a 102 ± 5 nm thick thermal oxide, and purchased from Siltronix, France. Before use, the wafers are cut into 1.5 cm × 1.5 cm slides and cleaned in a mixture of sulfuric acid and hydrogen peroxide in a 7 : 3 volume ratio at 130 °C for 10 min. Then the substrates are thoroughly rinsed in deionized water and dried by 99.99% nitrogen gas. This is followed by further cleaning in an oxygen plasma for 20 min. whereupon the slides are ready for use.

The polymers are polystyrene homopolymers purchased from Scientific Polymer Products (Ontario, NY), with weight-average molecular weights, M_w, of 2.4 and 44 kg/mol and polydispersity index less than 1.07. To make the sample films, we dissolve the as-purchased polymer in toluene with a concentration between 0.1 and 2 wt%. Then we filter the solution through a teflon filter with a nominal pore size of 0.1 μm before spin-coating it onto a cleaned substrate whereupon a thin film of the polymer is formed. The thickness of the polymer is controlled by controlling either the concentration of the polymer solution or the spinning speed, and measured by ellipsometry at 5 different locations of the film surface. Deviation in the film thickness is less than 0.3 nm. Unless otherwise stated, the films are not pre-annealed before measurement. To maintain the sample temperature during measurement, we placing the sample film on a temperature-controlled hot plate inside an air-tight chamber

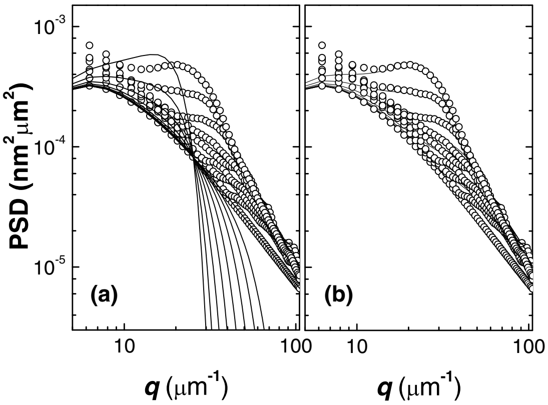

Fig. 2. Model fits (solid lines) by Eq. (2) (a) and Eq. (4) (b) to the power spectral density (PSD) (open circles) of a 4 nm thick polystyrene film dewetting from an oxide-coated silicon substrate upon annealing at 61 °C for different times. From bottom to top, the annealing times are: 0, 10, 20, 40, 80, 160, 320, 640 and 1250 s.

filled with 99.99% nitrogen at a pressure slightly above ambient. The temperature is monitored by a thermocouple thermally attached to the surface of a silicon wafer placed side-by-side the sample film.

2.2 Method to measure the dewetting dynamics of polymer films

It has been shown that polystyrene nanometer films supported by oxide-coated silicon are unstable [9] and dewet by the spinodal mechanism [10–13]. Conventionally, the height profile of the films upon dewetting is modelled by the theory of Vrij and Overbeek [20], which reads:

$$A_q^2(t) = A_q^2(0)\exp(2\Gamma(q)t), \tag{2}$$

$$\text{where} \qquad \Gamma(q) = -\frac{h^3q^2}{3\eta}[\gamma q^2 + G''(h)] = -\frac{\gamma h^3 q^2}{3\eta}(q^2 - q_c^2). \tag{3}$$

In the above $A_q^2(t)$ is the amplitude square of the q^{th} Fourier component of the height fluctuation of the film at time t and commonly known as the power spectral density (PSD), h is the film thickness, γ is the surface tension, η is the viscosity, and $G(h)$ is the van der Waals potential of the film. According to Eq. (3), $\Gamma(q)$ is negative when $q > q_c$ ($\equiv [-G''(h)/\gamma]^{\frac{1}{2}}$), but positive otherwise. Since $A_q^2(t) \sim \exp(2\Gamma(q)t)$, the height fluctuation (or surface capillary) modes with $q > q_c$ decay with time while those with $q < q_c$ grow exponentially. In addition, because $\Gamma(q)$ has a maximum at $q = q_c/\sqrt{2}$, the PSD grows the fastest at this wavevector, causing a characteristic length scale ($= 2\sqrt{2}\pi/q_c$) to develop in the morphology of the film as frequently noted in dewetting experiments [19,21,22].

To determine the PSD of a film, we capture the topographic image of the film by using tapping-mode atomic force microscopy (AFM). Then we multiply the data by a Welch function and subject it to Fourier transformation and radial averaging, which result in the PSD.

We have studied the evolution in the surface structure of a $h = 4$ nm film upon dewetting. The PSD acquired at different dewetting times is shown in Fig. 2(a) (open circles). The solid lines depict the best fit of Eq. (2) to the data. As seen, the model

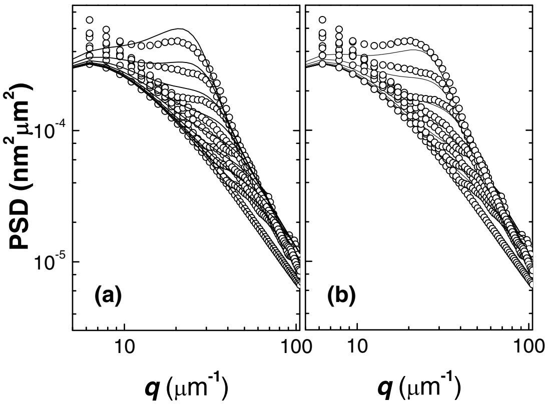

Fig. 3. Demonstration of how the quality of fit of Eq. (4) to the data depends on the value of η used in the model. The data are the same as those displayed in Fig. 2. The fitted lines shown in (a) and (b) are obtained by setting η to be 20% below and above the optimum fitted value, respectively. These fits are visibly inferior to those shown in Fig. 2(b), obtained by using the optimum fitted value of η.

does not fit the data at all. First, it predicts that the PSD at different times should cross each other at $q = q_c$ (where $\Gamma(q)$ changes sign), which is clearly not demonstrated by the data. Secondly, the PSD's do not show a preferred wavevector at which the mode amplitude grows the fastest. These show that some element crucial to the dewetting process is missing from the model.

We have previously shown by experiment [22,24] and computer simulation [22] that the films with $h \geq\sim 4\,\mathrm{nm}$ actually dewet by simultaneous spinodal and thermal nucleation mechanisms. Specifically, full-thickness holes were noted to nucleate in these films at random locations and random times [22,24], and the holes did not occur at the same locations when the experiment was repeated on a new film re-cast on the same substrate where the previous dewetted film had been washed away [22]. These evidence that the holes were due to nucleation induced by stochastic thermal noise. The occurrence of stochastic nucleation suggests that thermal noise is significant in the dewetting process. Therefore, we revise the model to account for thermal noise, by using the treatment previously employed in the counterpart problem of spinodal phase separation [19,26,27]:

$$A_q^2(t) = A_q^2(0) + \left[\frac{k_B T}{G''(h) + \gamma q^2} - A_q^2(0)\right](1 - \exp(2\Gamma(q)t)). \tag{4}$$

In the above, k_B is the Boltzmann constant and T is the dewetting temperature measured in degrees Kelvin; a time-independent backgound, $A_q^2(0)$, has been incoporated to allow for the fact that the film surface is not perfectly flat to begin with [25]. It should be remarked that eqn. 4, having been derived by linearization of the dynamic equations, is only applicable to data acquired in the early stage of dewetting before any holes are formed. We fit eqn. 4 to the data in Fig. 2(a), and display the fitting result in Fig. 2(b) (solid lines). As seen, the revised model fit the data noticeably better. In obtaining the fitted lines, we adjusted the viscosity of the film, η, only; the other model parameters were kept equal to the experimental or literature values as appropriate [23]. We find that the quality of the fit is quite sensitive to the value of η used in the model. Specifically, when the value of η is varied by 10% to 20% from the optimum fitted value, the quality of the fit deteriorates visibly as illustrated in Fig. 3.

This shows that our fitting scheme can provide a sensitive measure to the viscosity of the films.

3 Controversy between the dynamics and T_g of polymer films

Fukao and Miyamoto [28] were the first to study the anomalous dynamics of polystyrene films sandwiched between aluminum by using ac dielectric spectroscopy. (Notice that this is a different system from that studied by Keddie et al. [1], which was polystyrene on silicon.) These authors found that the dielectric loss peak of the films broadened systematically as the film thickness was decreased, demonstrating increased heterogeneity in the dynamics of the polymer upon confinement. On the other hand, the position of the peak did not change until the film thickness was decreased to about 10 nm whereupon the peak position shifted to a higher frequency or lower temperature, demonstrating enhancement in the modal mobility. Before this study [28], Forrest et al. [29] measured the T_g of a similar system, namely polystyrene sandwiched between silicon oxide. They found that the T_g of the films decreased with decreasing film thickness in much the same as that observed by Keddie et al. [1]. This result, showing that the T_g of capped polystyrene films also decreases with decreasing film thickness, is consistent with the finding of Fukao et al. [28] that the modal mobility of the films increases upon reduction of the film thickness.

However, this result was challenged by a later study of Sharp and Forrest who re-examined the T_g of capped polystyrene films [30], but this time on polystyrene sandwiched between aluminum, namely the same system studied by Fukao and Miyamoto [28]. Importantly, Sharp and Forrest adopted a different method in preparing the samples. In the previous experiments [28,29], the capping layer was fabricated by direct thermal evaporation onto the polymer. But in this experiment [30], the films were prepared by bringing two polystyrene half-films, spun-cast on aluminum-coated silicon, into contact and allowing them to integrate by surface forces. They found that the T_g of the films prepared by this method showed no thickness dependence, contrary to what Forrest et al. had found when the capping layer was fabricated by thermal evaporation [29]. This raises concerns about the effect of thermal evaporation on the dynamic properties of the polymer.

In light of the issue with thermal evaporation, Serghei et al. [13] re-examined the ac dielectric spectroscopy of polystyrene films, but this time with a different sample configuration. Specifically, they added micrometer-size spacers on top of the polymer film before adding the top electrode. They found that the data obtained by this setup exhibited no broadening or change in position of the dielectric loss peak, contrary to the result of Fukao et al. [28]. This result is also inconsistent with the observation that the T_g of these films decreased with decreasing film thickness [1]. However, it is consistent with an earlier result of Sharp et al. [12] who studied ac dielectric spectroscopy of isotatic PMMA on aluminum with a setup similar to that used by Serghei et al. [13], and also did not see any change in the width or position of the dielectric loss peak. But when they measured the T_g of the films by ellipsometry, they found that the T_g decreased substantially with decreasing film thickness. This result, and those discussed above on PS films show that the dynamic properties manifested by ac dielectric spectroscopy are not always consistent with the T_g measured by ellipsometry.

During thermal evaporation, the temperature of the evaporative source is increased until the source material evaporates. Since the interface of a metal and/or oxide film fabricated by thermal evaporation typically exhibits a nano-scale grainy structure [32], the evaporative particles from the source must have sizes no more than a few nanometers. When thermal evaporation is carried out on a soft polymer,

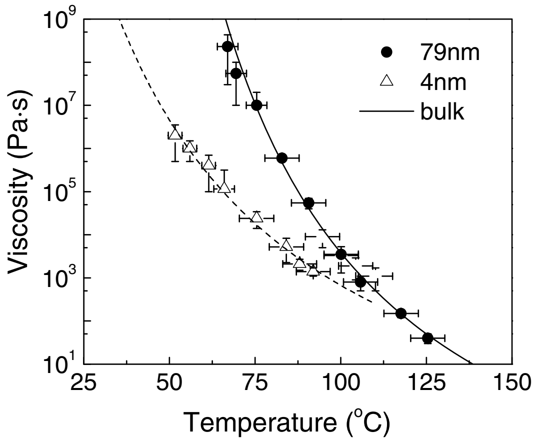

Fig. 4. Viscosity vs. temperature of polystyrene films ($M_w = 2.4$ kg/mol) on oxide-coated silicon with thickness, $h = 4$ and 79 nm. The solid line represents the published viscosity of the bulk polymer [17]. The dashed line is the best fit to the VFT relation (Eq. (1)).

the evaporative particles could be energetic enough to hammer the top region of the polymer into a nano-porous structure whereby the interfacial area of the polymer is substantially increased. In fact, the influence of interfacial effect on the dynamics of polymers had been successfully observed by Blum et al. using nuclear magnetic resonance [31]. In this experiment [31], the samples were thin films of polymer adsorbed on high-surface-area fume particles. The importance of having a large interfacial area is apparent. We thus feel that lacking of sensitivity cannot be excluded as a possible cause for the absence of dynamic anomalies found in some experiments. In a recent ac dielectric spectroscopic measurement by Serghei et al. on a $h = 12$ nm PS film [13], the scatter in the data amounts to $\sim 1/5$ the height of the main loss peak (Fig. 4 of Ref. [13]). Since the dynamic heterogeneity of PS films has been found to exist within a surface mobile layer with thickness $\leq \sim 2$ nm [33], the dynamic anomaly arising would be barely visible. On the other hand, Reiter [34] had found that PS films with thickness below 8 nm supported by a glass substrate dewetted visibly at temperatures below the bulk T_g. This implies that we may be able to detect the non-bulk-like dynamics of PS supported films by studying how they dewet near the bulk T_g. Below we show the results obtained by studying the dewetting of PS films supported by oxide-coated silicon with thicknesses from 2.3 to 79 nm.

4 Results and discussions

Figure 4 shows the viscosity of polystyrene films with $M_w = 2.4$ kg/mol and thickness, $h = 4$ and 79 nm plotted versus temperature. The published viscosity of the bulk polymer [17] is shown by the solid line. Figure 5 shows the $T_g(h)$ of similarly prepared polystyrene films obtained by ellipsometry, which is verified to resemble the data of Keddie et al. [1]. From Fig. 5, the T_g of the films becomes bulk-like when the film thickness is $>\sim 100$ nm, consistent with the data shown in Fig. 4 that the viscosity of the 79 nm films overlap with that of the bulk. On the other hand, the viscosity of the 4 nm films is substantially lower than that of the 79 nm films for all the temperatures studied. Moreover, it diverges much more slowly with decreasing temperature. We find that the data fit well to the VFT relation (Eq. (1)) if the fitted value of T_K is reduced by 39 °C from the bulk value but those of τ_∞ and A are kept the same as the

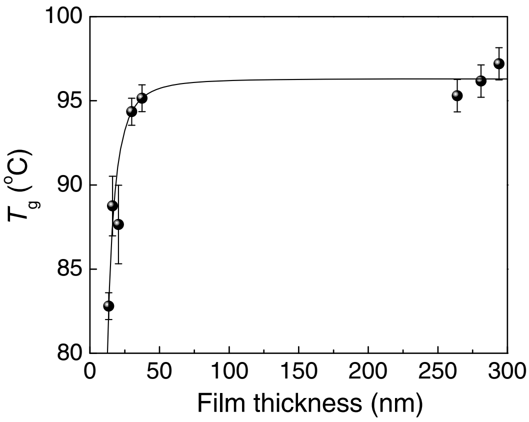

Fig. 5. A plot of T_g vs. film thickness of polystyrene ($M_w = 44\,\mathrm{kg/mol}$) spun-cast on oxide-coated silicon. The films were pre-annealed in a 10^{-2} torr oil-free vacuum oven at $150\,^\circ\mathrm{C}$ for 1.5 h before the measurement, which was carried out in air during a cooling scan upon soaking the sample film at $150\,^\circ\mathrm{C}$ for 20 min. The solid line is the best fitted line to the model proposed by Keddie et al., i.e., T_g (in $^\circ$C) $= T_{g,\infty}[1-(h_0/h)^\delta]-273$, where $T_{g,\infty} = 373.8\,\mathrm{K}$, $h_0 = 3.2\,\mathrm{nm}$ and $\delta = 1.8$ [1]. In here, we find that $T_{g,\infty} = 369.5$ K, $h_0 = 3.4$ nm and $\delta = 2.4$.

bulk. This finding is in keeping with the discussion of Sec. 1 that a reduction in T_g may result from a reduction in T_K. The amount of shift in T_K found here is also in good quantitative agreement with that reported by Herminghaus et al. [35] who had deduced it from the growth rate of the fastest unstable capillary mode. Note that in here, T_K is deduced by fitting the PSD data to Eq. (4). Given the result shown in Fig. 2, it is not obvious that the two approaches would give the same result. Another difference between this work and Ref. [35] is that here, we investigate whether the viscosity of the films changes with the film thickness and the changes in viscosity, if any, are commensurate with the observed shift in T_g. But in Ref. [35], the focus was on the shift in T_g and its theoretical explanation; no value of viscosity or dewetting time was reported.

We have also studied polystyrene films with other thicknesses. Figure 6 shows the AFM topographic images taken from four films with thickness, $h = 2.3$, 5, 9 and 79 nm and annealing temperature, $T = 55\,^\circ\mathrm{C}$ for the 2.3 nm film and $70\,^\circ\mathrm{C}$ for the other three films. As seen, except for the 79 nm film, all the other films become visibly roughened within 4 hr of annealing. In addition, the thinner the film, the sooner this occurs. To get an idea of how the viscosity of the films compare to the bulk, we calculate $\Gamma_{\max}(\eta_{\mathrm{bulk}})$, the growth rate of the fastest unstable capillary mode discussed in Section 2.2 assuming the viscosity of the film to be the same as the bulk. In the first row of Table 1, we enumerate the values of $[\Gamma_{\max}(\eta_{\mathrm{bulk}})]^{-1}$ we found for the films, which may be taken as an estimate of the theoretical dewetting time of the films if the film viscosity were the same as the bulk. In the second row of Table 1, we enumerate the experimental times when the films were first found to be visibly roughened, which may be taken as an estimate for (the upper limit of) the experimental dewetting time. In the last row of Table 1, we enumerate the ratio of the theoretical to the experimental dewetting time. As seen, the experimental dewetting time is at least 1150 times shorter than the corresponding theoretical time. Moreover, the thinner the film, the bigger the discrepancy is. This result shows that the viscosities of all three films are smaller than the bulk. Moreover, the thinner the film, the smaller the viscosity is, consistent with the thickness dependence of the T_g measured by ellipsometry.

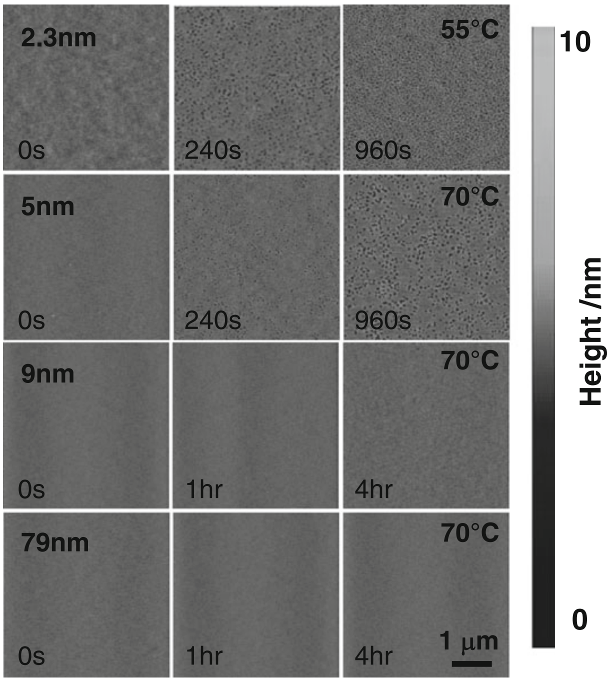

Fig. 6. Atomic force microscopic topographic images of polystyrene films ($M_w = 2.4$ kg/mol) with thicknesses of 2.3, 5, 9 and 79 nm undergoing dewetting at different temperatures and dewetting times as indicated.

Table 1. Comparison between the theoretical dewetting time ($[\Gamma_{\max}(\eta_{\text{bulk}})]^{-1}$) and experimental dewetting time (τ_{roughen}).

h T	2.3 nm 55 °C	5 nm 70 °C	9 nm 70 °C
$[\Gamma_{\max}(\eta_{\text{bulk}})]^{-1}$	2.1×10^7 h	2000 h	4600 h
τ_{roughen}	240 s	240 s	4 h
$[\Gamma_{\max}(\eta_{\text{bulk}})]^{-1}/\tau_{\text{roughen}}$	3×10^8	3×10^4	1150

5 Conclusion

In summary, we have addressed the controversy concerning whether the dynamics of polystyrene films on silicon is consistent with their T_g. To measure the dynamics of the films, we developed a method to monitor the dynamics of surface capillary waves on the films from which the viscosity can be model-fitted. We used this method to determine the viscosity-temperature dependence of polystyrene films with thicknesses of 4 and 79 nm. We found that the viscosity of the 4 nm films was substantially smaller than that of the 79 nm films, and it had a much weaker temperature dependence. By comparing the dewetting times of several films with different thicknesses below 10 nm, we found that the experimental dewetting times were at least 1150 times shorter than the corresponding theoretical dewetting times assuming the viscosity of the films to be the same as the bulk. Moreover, the thinner the film, the bigger the discrepancy was. Altogether, our result shows that the viscosity of polystyrene films supported by oxide-coated silicon decreases with decreasing thickness, consistent with the thickness dependence of their T_g determined by ellipsometry.

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